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Methods for forecasting the effect of exogenous risks on stock markets

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ABSTRACT

Markets are subjected to both endogenous and exogenous risks that have caused disruptions to financial and economic markets around the globe, leading eventually to fast stock market declines. In the past, markets have recovered after any economic disruption. On this basis, we focus on the outbreak of COVID-19 as a case study of an exogenous risk and analyze its impact on the Standard and Poor's 500 (S&P500) index. We assumed that the S&P500 index reaches a minimum before rising again in the not-too-distant future. Here we present a forecast model of the S&P500 index based on the breaking news and publicly available information. We assumed that the biggest fall of the S&P500 during the COVID-19 outbreak will occur when the largest daily number of deaths was confirmed. We inferred that the peak number of deaths occurs 2-months since the first confirmed case was reported in the USA based on previous COVID-19 situation reports from other countries. We also compare the S&P500 and the DAX market dynamics around the COVID-19 crisis as well as other previous crises, demonstrating that the impact of market news is highly consistent across these multiple market crises. The forecast is a projection of a prediction with stochastic fluctuations described by q-gaussian diffusion process with three spatio-temporal regimes. Our forecast was made on the premise that any market response can be decomposed into an overall deterministic trend and a stochastic term. The prediction was based on the deterministic part and for this case study is approximated by the extrapolation of the S&P500 data trend in the initial stages of the outbreak. The stochastic fluctuations have the same structure as the one derived from the past 24 years. A reasonable forecast was achieved with 85% of accuracy.

1. Introduction

In the investigation of the stock market dynamics, there are two well defined types of models; the "descriptive" and "structural" which reproduce stock market dynamics [1]. Nowadays, these models attempt to capture the *endogenous* and *exogenous* systemic risks of market movements [1–3]. *Endogenous* risks have been modeled in a microeconomic fashion by considering the interactions between agents and how these change over time, causing non-linear disruptions even if system parameters change in a smooth fashion [4,5]. New "descriptive models" present *endogenous* systemic risks as a

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dynamic factor affected by the flow of market orders and time scale [2], excessive profits, and excessive losses [6] and large positive or negative variation in stock markets index [7] due to market activity. The recent "structural models" which are based on physical model systems such as a combination of oscillation within a basin of free energy [3], spin glasses [8], and kinetic Ising model to model stock market network [9] proposed a natural non-equilibrium system. In most cases, they introduce the *exogenous* systemic risks as a global risk component which evolves all the time affecting prices of traded assets. The effects of these systemic risks are likely causes of abrupt changes at the macro-level of market dynamics [10,11].

In both systemic risks (*endogenous* and *exogenous*) the market response can be decomposed into two parts, a response function that models changes in the overall trend of the system, called the skeleton [12], and the other is the stochastic term. In this work, we analyze the nonlinear properties of both of these terms.

The *endogenous* systemic risks are an inherent part of the nonlinear dynamics of a market and may have detectable precursor signals that act as warnings similar to those used in other nonlinear systems such as climate and ecology [13]. As an example, the 1987 market crash is likely an endogenous event [14], as it had a measurably different effect on market dynamics than the September 11 attacks [15] and the 1995 Kobe earthquake [14] which pose systemic *exogenous* risks to markets. The last two events cannot be endogenized into market prices by 'rational' agents ahead of time because they are not foreseeable [16].

This paper presents the solution of a critical outstanding problem in finding how the market is coupled to the globe, and how exogenous and unpredictable global events produce deterministic trends in the market. In Section 2, we present the model for the forecasting of S&P500 index's response due to the outbreak of COVID-19. In Section 3, we compared the dynamics around the COVID-19 crisis with other previous crises for the S&P500 and the DAX. In Sections 4 and 5 , the deterministic trend and the stochastic noise are developed for the forecasting of S&P500 index's response due to the outbreak of COVID-19. In Section 6, we present a discussion on how this model provides an overview of the impacts of an exogenous market shock that can be used as a reference in developing predictive models with more accuracy.

2. The model

Our model is based on an empirical analysis of previous financial crises. For a specific financial crisis, our forecasting method uses breaking news as an input to define when this particular economic crisis starts and its corresponding recovery occurs. In this paper we are focusing on the development of a forecast for the fluctuations of the S&P500 due to COVID-19 as a case study.

For this analysis, it is assumed that COVID-19 poses systematic *exogenous* risks that affect stock markets behavior. Therefore, our forecast model considered epidemiological research results as news breaks to project the impact of COVID-19. Epidemiological researchers from around the world have produced an extensive array of analyses in order to model the spread, growth, peak, and ultimately the decline of the disease [17–21]. For countries like China, Japan, Italy, and Iran, their epidemiological curve of COVID-19 progression illustrate a peak before the second month since the first cases were detected [22–24]. In countries like the UK, Australia and Germany, the governments have taken mitigation measures to slow the impact [17,18]. The simultaneous reaction of governments, companies, consumers and media, have created a demand and supply shock, making COVID-19 a qualitatively different economic crisis than previous crises [25]. This economic 'wedging' of falling supply and demand is caused by rapidly escalating unemployment decreasing consumer demand where, in the US for example, 17 million Americans have applied for unemployment insurance in the first three weeks of the crisis [26], and businesses are closing their doors reducing the supply of manufactured goods across the globe [27]. As a consequence financial markets around the world have fallen precipitously and market volatility is at near-all-time highs [28]. For example the S&P500 has registered the worst one-day fall in the last 24 years and the third biggest percentage loss in its history.

In the uncertain environment of COVID-19 it is difficult to forecast the fluctuations of the S&P500. We then need assumptions such as the mortality rates due to COVID-19 or the duration of the current shutdown of economic activity, both are considered *publicly available information* due to their wide distribution to the general public. Our central assumption is based on a predicted peak for COVID-19 deaths. Current results in Fig. 1 based on the daily World Health Organization reports [29] show that the S&P500 responds to the inflection points of the cumulative amount of deaths and confirmed cases with no lag. This fact supports our assumption that when a peak number of deaths occurs, the market reaches a stationary point.

We have presumed that the day with the largest number of confirmed deaths in the USA will be the day with the largest drop in the S&P500 index. This day was assumed to occur 2-months since the first COVID-19 confirmed case occurred [30]. Once the minimal stock market price is achieved the recovery period begins and continues as economic activity starts to return gradually to normal. To construct the forecast, we assume that the stock market index can be decomposed into a deterministic trend and a stationary stochastic fluctuation. The statistics of the fluctuation have been obtained by analyzing the S&P500 index during the 24 year period from January 1996 to March 2020 [31]. $I(t_0)$ is the initial point of the stock market index for some time point t_0, and the index $I(t)$ for $t > t_0$ is its time evolution. In these predictions we take $t_0 = 24/03/2020$ and $t_0 = 28/02/2020$ for the two different forecast. The price return at time t is defined by:

$$X(t) = I(t_0 + t) - I(t_0). \tag{1}$$

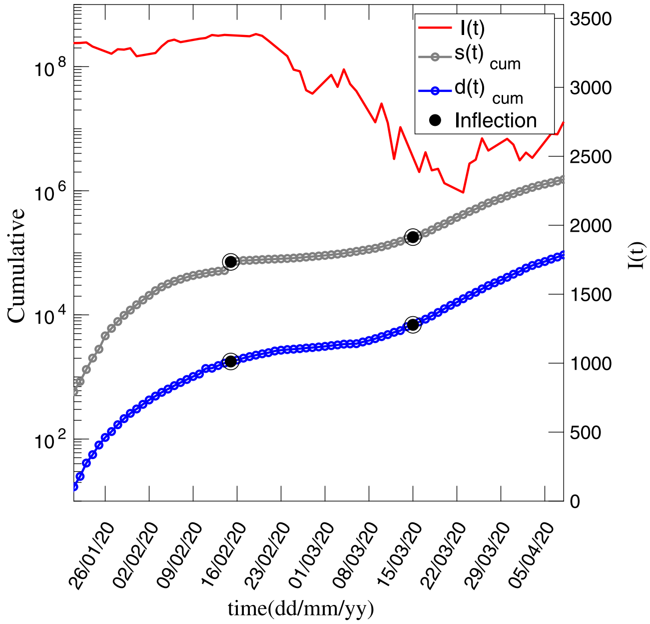

Fig. 1. The S&P500 decreased after the WHO (World Health Organization) reported an increment in confirmed cases $s(t)$ and number of deaths $d(t)$ due to COVID-19. The increments in the number of cases are represented with a black filled point, which are an abrupt change of slope of the first derivative of $s(t)_{cum}$ and $d(t)_{cum}$.

In earlier work we used a detrended fluctuation approach [31] to decompose the price return $X(t)$ into a deterministic component $\overline{X}(t)$ and a stationary *fluctuating component* $x(t)$

$$X(t) = \overline{X}(t) + x(t). \tag{2}$$

The trend $\overline{X}(t)$ was obtained by averaging the index over a moving time window. The size of the window was one year, that was optimized to guarantee that the fluctuations around the trend exhibit stationary behavior. The governing equation of this stationary behavior is [32]:

$$t^{1-\xi}\frac{\partial p}{\partial t} = \xi D^{\xi}\frac{\partial^2 p^{2-q}}{\partial x^2}, \tag{3}$$

By using a curve-fitting analysis on the S&P500 data over the past 24 years (see Appendix A), we show that the probability density function (PDF) of the detrended price is well described by the functional form:

$$p(x,t) = \frac{1}{(Dt)^{1/\alpha}} g_q\left(\frac{x}{(Dt)^{1/\alpha}}\right), \tag{4}$$

where $\alpha = \frac{3-q}{\xi}$. Being D, q and α time-dependent fitting exponents which analysis is shown in Figure A.1. The g_q term is the q-Gaussian function defined as:

$$g_q(x) = \frac{1}{C_q} e_q(-x^2) \tag{5}$$

and the q-exponential function is $e_q(x) = [1+(1-q)x]^{\frac{1}{1-q}}$ that reduces to the exponential function when $q \to 1$. The normalization constant C_q for $1 < q < 3$ is given by:

$$Cq = \sqrt{\frac{\pi}{q-1}}\frac{\Gamma((3-q)/(2(q-1)))}{\Gamma(1/(q-1))}. \tag{6}$$

The cumulative distribution function (CDF) of the PDF (Eq. (4)) of the detrended price return is;

$$F(x,t) = \int_{-\infty}^{x} p(x,t)\,dx. \tag{7}$$

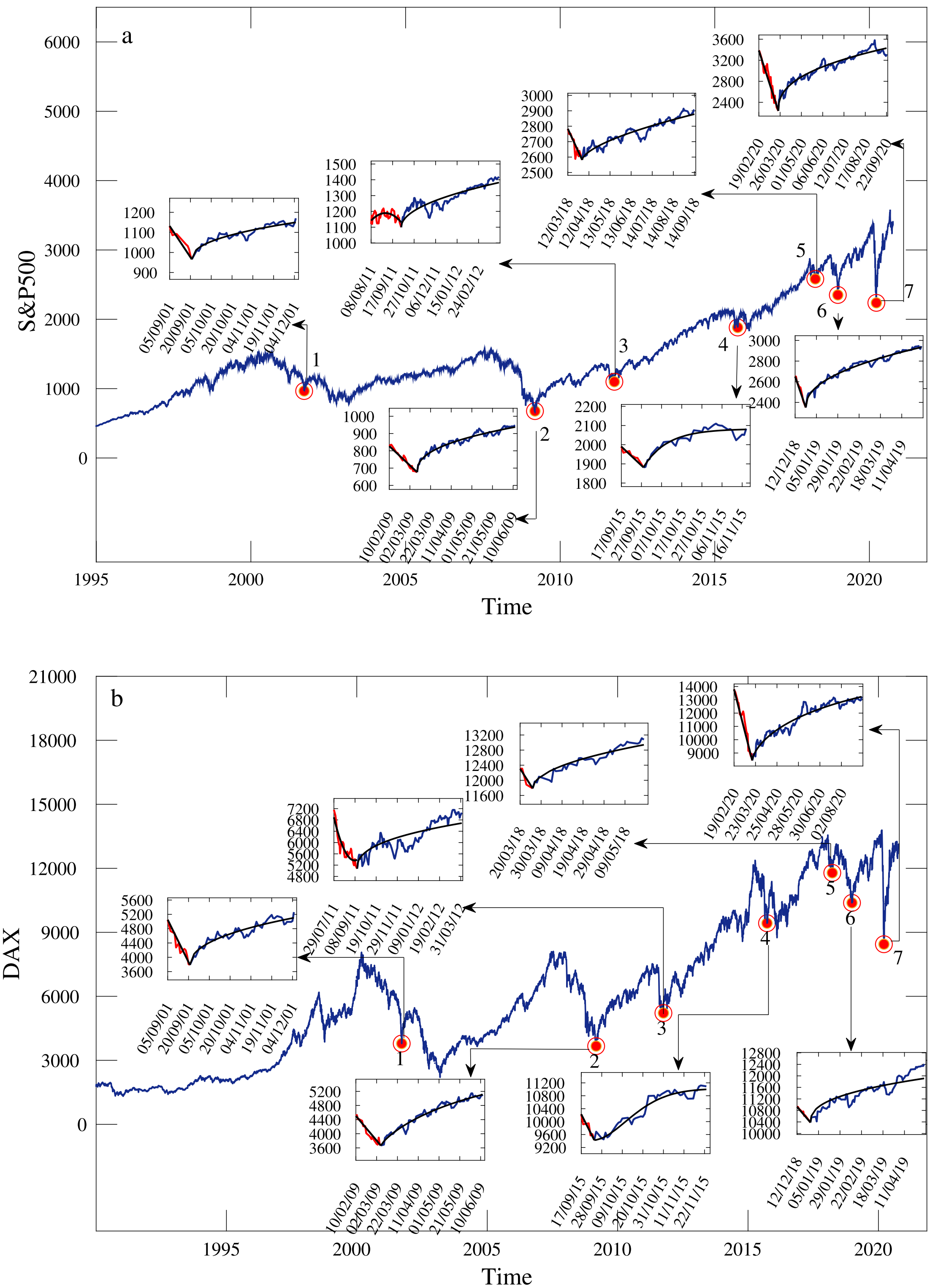

Fig. 2. (a) The S&P500 index $I(t)$ from 03/01/95 to 06/10/20 (25 years). In this data set the largest daily percentage loss of −11.98% was registered on 16/03/20. Some other key events are also shown for reference. The description of each crisis is in Table 1. (b) The DAX index from 03/01/92 to 06/10/20 (28 years). In this data set the main financial crises were fitted by applying Eq. (11). The largest daily percentage loss of DAX was −11.94%. Some other key events are also shown for reference.

The complementary of the CDF is used here to quantify risk under extreme events [33]. This is defined as probability that $X - \overline{X}$ lies outside the interval $[-x, x]$ is

$$P(X(t) - \overline{X}(t) > |x|) = 1 - \int_{-x}^{x} p(x,t)\, dx. \tag{8}$$

For this paper we introduce the standardization of the q-error function

$$\mathrm{erf}_q(x) = 2\int_0^x g_q(y)dy \tag{9}$$

The standard error function is a special case of the q-error function for $q = 1$. Considering this definition, Eq. (8) is written as:

$$P(X(t) - \overline{X}(t) > |x|) = 1 - \mathrm{erf}_q(x/(Dt)^{1/\alpha}). \tag{10}$$

We use Eq. (10) to develop a forecast of the market based on the historical fluctuations in the S&P500. Fig. 2 shows some key dates over the last 24 years of the S&P500. The market dynamics consists of periods of a bull market (systematic increase of the index) and a bear market (systematic decrease). The crashes in this plot are discontinuous. They occur in a very short period of time that we attribute as noise that it is added to the deterministic component. For example, during the Global Financial Crisis (GFC) there is a long-term decline in the value of the S&P500 that we can interpret as the deterministic trend, but near the point where Lehman Brothers collapses there is a discontinuous market crash. In fact, there are several very large daily price movements that are not always attributable to a specific event. These 'crashes' (both up and down) are more similar to large stochastic movements, not a part of the deterministic component of the market dynamics.

3. Analysis of previous financial crises

We performed an empirical analysis of seven financial crisis using the American S&P500 and European DAX indexes, as shown in Fig. 2. From this figure, we inferred that the lowest registered price in DAX usually occurs within a margin of three single-days average once the lowest price has been achieved by S&P500. Additionally, we find an empirical relationship between breaking news and transition between collapse and recovery in stock markets. We applied this empirical correlation in our forecast method for the stock market crash due to COVID-19. We note that any forecast's accuracy depends on the proper identification of breaking news [34–43] and the corresponding response of the authority, entity, or company involved and affected [44–52].

The method to construct the expected recovery trend once the price has dropped sufficiently low is based on a hypothetical correlation with publicly available information. To prove our hypothesis, we have analyzed two different indexes: the S&P500 from the USA and the DAX from Germany, both of them have been analyzed considering approximately the last 25 years. Seven events classified as *"crashes"* or financial crises have been detected and analyzed separately. Table 1 contains the analyzed financial crises. Then, it has been noticed that each financial crisis is related to breaking news. The breaking news is classified as crash or recovery news. The crash news produces a systematic decrease in the stock market price because it creates uncertainty in traders who face an exogenous event that was not foreseeable. On the other hand, the recovery news produces a systematic increase because the uncertainty seems to be under control once the corresponding response of the authority, entity, or company affected is known.

4. Deterministic trend

To establish the effect of COVID-19 on the market we assume that it has a deterministic, exogenous impact. This impact is assumed to be a response function that corresponds to a COVID-19 induced a collapse in the market followed by a short transition called expected recovery. As a further refinement we assume that this transition is dictated by a key quantitative factor: the estimated date on which the mortality rate peaks. Our first estimation of this trend (a private communication between authors on April 2, 2020) was based on the time-frame from 31/01/2020 to 29/02/2020 in which a stock market decline in $I(t)$ had already been observed. Initially, for the construction of the deterministic trend $\bar{I}(t)$ two functions were used: parabolic and hyperbolic. After analyzing previous crises, a piecewise function consisting of a linear function and an exponential function for the collapse and recovery of the market was used.

Based on the time estimations of the expected recovery, these three functions were constructed by applying different procedures. For the parabolic function, three conditions were applied and are illustrated in Fig. 3. First, the initial point of the predicted trend is $(t_0, \bar{I}_{t_0})$, where $t_0 = 29/02/2020$. Second, the slope of $\bar{I}(t)$ at t_0 is obtained from a linear fitting within the time interval of 31/01/2020 to 29/02/2020. Third, the point where the recovery is expected to occur is t_f. For this case t_f was located 2 months since the first confirmed case was reported in USA, the 23/01/2020 by WHO [53]. The recovery point, t_f, is a local minimum. For the hyperbola three similar conditions were applied: The slope of $\bar{I}(t)$ at t_0 is the same as the parabola, the slope of the expected market recovery is assumed to be 0.5 of the slope of the market's collapse, similar to the previous crashes that have occurred over the past 25-years, and the intersection of the two slopes occurs at the recovery point t_f. The hyperbola proved to be an acceptable fitting for the initial recovery stage. The parabola

Table 1
Analysis of the largest exogenous financial crises.

Nro	Event	Crash-New (CN)	Recovery-New (RN)	Days
1	World trade center attacks	11/09/2001: The September 11 attack is confirmed as a terrorism attack [34].	21/09/2001: George W. Bush, US President, declares the war to Afghanistan by sending the American forces to fight the Talibans [44].	11
2	Bottom of Great Recession	13/02/2009: After the collapse of Lehman Brothers' company, the US Senate rejected the proposal of the US House to approved $787 billion stimulus package [35].	12/03/2009: B. Obama, U.S president, signed the stimulus package [45]. Then in a public conference he ensured that he will reassert fiscal discipline and financial regulation to intervene cross-border financial transaction [46].	28
3	U.S removed as a free-risker borrower	21/09/2011: Standard & Poor's removed the United States government from its list of risk-free borrowers for the first time [36]. Then, Moody's Downgrades Credit Ratings of Three Large Banks [37].	04/10/2011: Due to European financial crises, the dollar gained in value against the Euro and the U.S [47]. Treasuries turned a profit [48].	14
4	Chinese Market Turbulence	16/09/2015: The confirmation of the Chinese market crash [38]. The Chinese National Development and Reform Commission advised to issue more bonds and borrow more loans in offshore markets during the crisis [39]. This creates speculation on the Yuan's currency.	23/09/2015: Xi Jinping, China President, arrived to U.S. to look for a strategic relationship and informed that the "government took steps to stabilize the market and contain panic in the stock market and this avoided systemic risk."[49]	8
5	Trumps plans to impose China $ 60 billion in tariffs	22/03/2018 The US administration unveiled tariffs designed to retaliate against China for intellectual property breaches , imposing about $60 billion in retaliatory charges [40]. China responded by threatening to impose tariff on steel and aluminum imports [41].	02/04/2018: Commerce Secretary of US, Wilbur Ross, says China's new tariffs do not represent a threat to the United States [50].	12
6	Trumps attempts to use US treasury	04/12/2018: Markets tumbled after Trump tweeted "I am a Tariff Man" and the Trump administration backed off earlier claims of a trade-war truce with China [42].	24/12/2018: US secretary treasure, Steven Mnuchin, calmed markets by informing that the Federal Reserve will no be touched [51].	21
7	COVID-19 outbreak	29/02/2020: The first confirmed death in US is reported, Trump imposed travel restrictions to China, Iran, South Korea and Italy [43].	23/03/2020: The biotechnology company Moderna said its experimental vaccine for Covid-19 could be available to a select few as soon as next fall, ahead of expectations for a commercial release in a year [52].	24

did not show a good fitting and it is probably more suitable for cases where the crisis leads to prolonged transitions such as the bear–bull market.

The proposed piecewise function was constructed based on an empirical analysis of past financial crises in S&P500 and DAX as seen in Fig. 2. The analysis demonstrates that the collapse and recovery in different stock market indexes are highly consistent across multiple market crises. The piecewise function is defined by a linear and exponential function, as illustrated in Fig. 4a. The linear function is obtained from a straight line between t_0 and t_f, these points are related to Crash-News (CN) and Recovery-News (RN), respectively. The exponential function is used to approximate the recovery period after t_f. The piecewise function is defined as:

$$\begin{aligned} \bar{I}(t) &= \left(\frac{\bar{I}_f - \bar{I}_{t_0}}{t_f - t_0}\right)(t - t_f) + \bar{I}_f && \text{if} \quad t < t_f \\ \bar{I}(t) &= \bar{I}_f + (\bar{I}_\infty - \bar{I}_f)\left[1 - e^{-\left(\frac{t - t_f}{bt_c}\right)^c}\right] && \text{if} \quad t > t_f \end{aligned} \tag{11}$$

where, $\bar{I}_{t_0}$ is the index price at the initial point, t_0, of the predicted trend. The variable $\bar{I}_f$ represents the lowest index price registered during a particular collapse, $\bar{I}_\infty$ is a constant value, and $t_c = 1$ day. To fit the dimensionless parameters

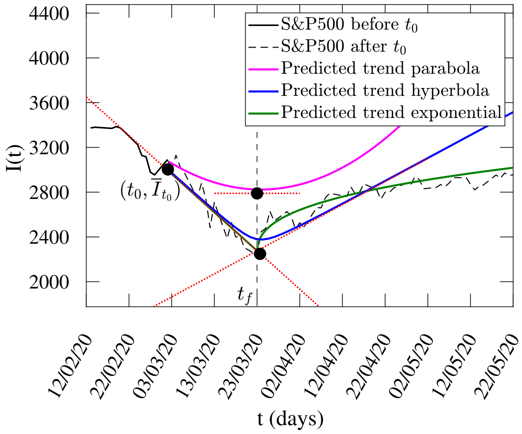

Fig. 3. A downward trend followed by a market recovery is estimated by applying a parabolic fitting (magenta line), hyperbolic fitting (blue line) and piecewise fitting (green line). Some conditions were applying to define the parabolic and hyperbolic deterministic trend. The time when the prediction starts is t_0. Then, the slope of the collapse is calculated within a time interval of 31/01/2020 to 29/02/2020. We assumed that there is no time lag between the peak of mortality rate and the time where the market starts to recover. Finally, the recovery is gradually and starts after the lowest market point t_f. A different process was applied for the construction of the piecewise function.

a, b, and c, we normalize the trend as $\widehat{I}(t) = (\bar{I}(t) - \bar{I}_f)/\bar{I}_f$. From Eq. (11) we obtain:

$$\begin{aligned} \widehat{I}(t) &= \frac{\bar{I}_f - \bar{I}_{t_0}}{I_f}\left(\frac{t-t_f}{t_f-t_0}\right) && \text{if} \quad t < t_f \\ \widehat{I}(t) &= a\left[1 - \exp\left(-\left(\frac{t-t_f}{bt_c}\right)^c\right)\right] && \text{if} \quad t > t_f, \end{aligned} \tag{12}$$

where, $a = (\bar{I}_\infty - \bar{I}_f)/\bar{I}_f$. Each financial market recovery was fitted with Eq. (12). We found that the fitting parameters can be expressed in terms of the *magnitude* of the crisis, that is defined as;

$$m_f = \log_{10}\left(\frac{\bar{I}_f - \bar{I}_{t_0}}{\bar{I}_{t_0}}\right), \tag{13}$$

This variable is the logarithmic of the stock market percentage loss. We obtained empirical correlations between m_f and the fitting parameters a, b, and c that appear in Eq. (12). The results of these empirical relationship are shown in Figs. 4b-d. The largest percentage loss of the S&P500 history occurred during the COVID-19 outbreak; therefore, an extrapolation is made to approximate the corresponding coefficients using the empirical relationship previously obtained. Finally, the exponential function defined as Eq. (11) is constructed with the extrapolated values and is showed in the Fig. 4a. It is remarkable that these parameters tends to be *universal*, i.e. they are the same for all the crisis, under the condition that the crisis produces a market drop above 10%. For smaller crisis, the exponential relation with the magnitude are required to estimate the trends of the recovery.

5. Stochastic fluctuations

Up to this point we have evaluated the systemic risk as the deterministic aspect of the market evolution, which is often neglected by analysts in the determination of the impact of exogenous events such as the COVID-19 pandemic. In what follows we evaluate the risk associated with the stochastic fluctuations of the index by 'dressing' the deterministic risk with a q-Gaussian diffusion process. In a previous publication [32] we found that the q-Gaussian fluctuations of the S&P500 index around the trend have distinctive super-diffusion spatio-temporal regimes ('space' refers to the size of the fluctuations). To allow long-term forecasting on the order of days, we have extended the analysis for longer times. Three well-defined regimes are observed in Fig. 5,

- Zone A: Strong superdiffusion process with short-time correlations
- Zone B: Weak superdiffusion process with weak time correlations
- Zone C: Normal diffusion process with no time correlations.

These zones are well-distinguished in time, where Zone A goes from 0 to 38 min, Zone B from 10 days to 28 days and after a crossover period Zone C starts at 30 days (Fig. 5). Note that the time series tends to a classical diffusion process

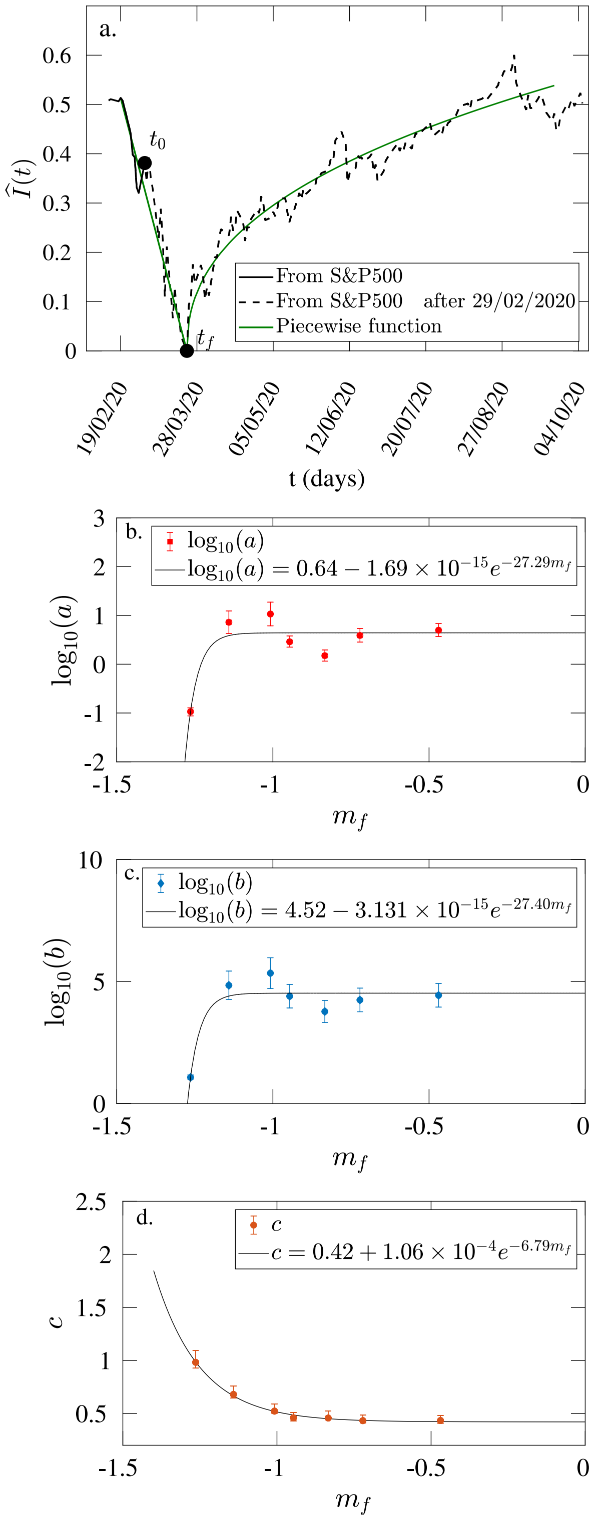

Fig. 4. (a) The decline and recovery on the S&P500 trend due to COVID-19 is observed. The trend was constructed by applying a piecewise function, where the linear function is associated with the collapse and the exponential function with the expected recovery. (b) The empirical correlation between coefficient a vs. m_f for previous financial crises. (c) The empirical correlations between coefficient b vs. m_f, and (d) the empirical correlations between coefficient c vs. m_f for previous financial crises. The piecewise function with $t_0 = 29/02/2020$ shows the most accurate result.

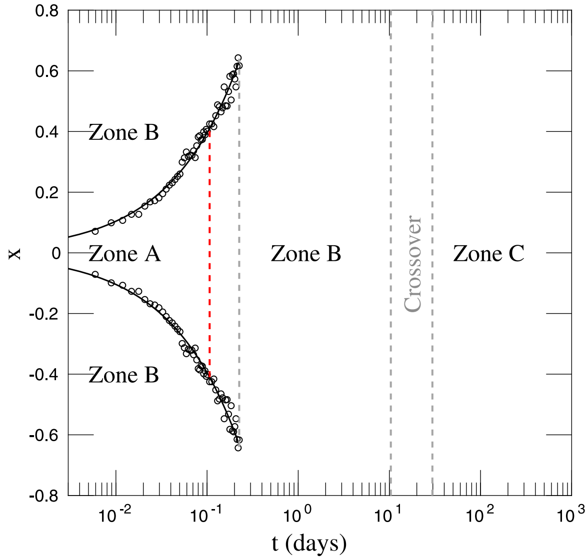

Fig. 5. A floor plan of the change of the parameters α and q in the time evolution of S&P500 PDFs'. Refer to Appendix A to find the exact values of q and α per each zone. The α and q values were obtained after fitting the PDF of price return of S&P500 data of the past 25 years with Eq. (4). Three different zones were determined. The circles represent the end points of the strong super-diffusion regime (zone A) from t_0 to $t = 35$ min. The remaining area during the first 35 min to 28 days corresponds to a weak super-diffusion regime (zone B). A normal diffusion process is reach after thirty days (Zone C). The gray dashed lines represent the transitions between Zone B and Zone C.

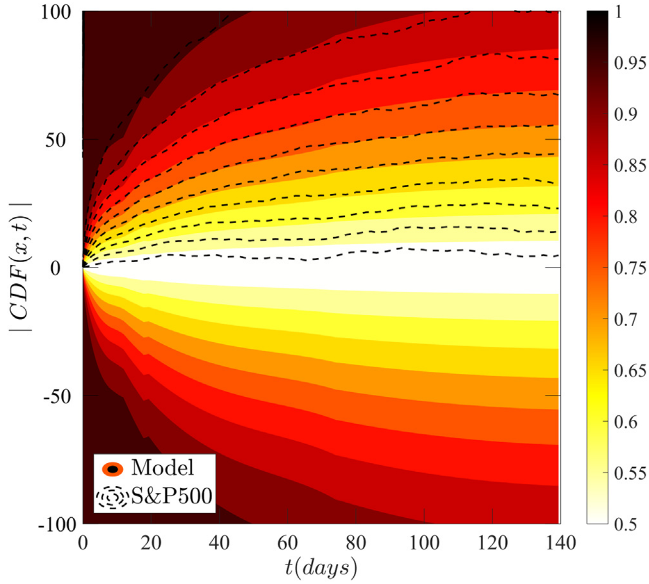

Fig. 6. Comparison of time evolutions of *CDFs* between the price return of detrended stock market data and the analytical model which considers the time-dependent fitting parameter obtained from the second technique.

for large times, in agreement with the classical Central Limit Theorem. See [32] for a complete analysis of these zones and their derivation.

Two techniques were used to calculated the optimal parameter q for each distribution, one focusing on the PDF; and the other focusing on each CDF of detrended price return. For the first technique two methods were used. The least square method and q-moments method. The least square method applies Eq. (4) as the fitting function. The q-moments method uses a system of two equations and two variables, q and α. The first equation is given by the "*second moment*" or variance $\langle x^2 \rangle$. The second equation refers to the "*escort second moment*" or q-variance $\langle x^2 \rangle_{q_2}$. In general, the q-moments are applied to PDFs with asymptotic decays because they provide finite values [54]. The second technique is based on each CDF, where a least squares method is applied using Eq. (A7). The quality of the fitting models was evaluated after reproducing the time

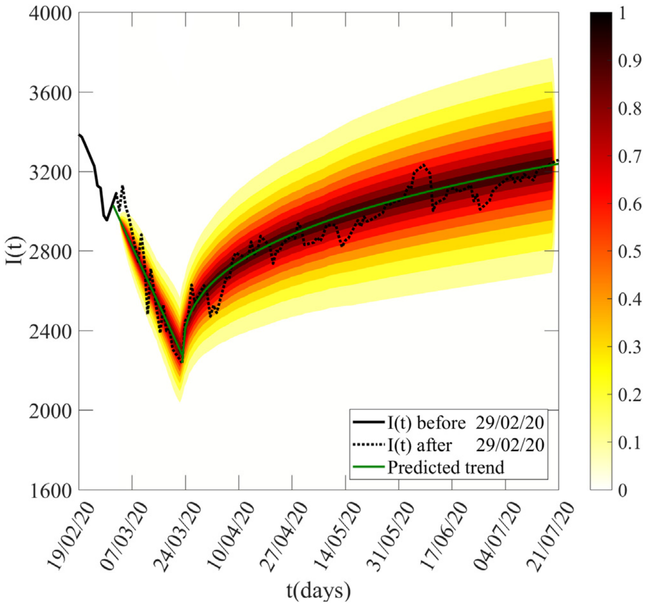

Fig. 7. The decline and recovery on $I(t)$ due to COVID-19 are observed from 29/02/20 to the following 5 months. A downward and upward market predicted trend is calculated by applying a piecewise function, which consists of a linear and an exponential function. The trend was constructed with no time lag. The uncertainty is represented by a scale from 0 to 1, where represent the $P(X - \overline{X} > |x|)$ of each contour line. The economy will start its recovery after this epidemic peak of deaths.

evolution of $CDF(x, t)$ with the time-dependent fitting parameters previously obtained, then we compare them with the *CDF* of detrended price return. The best results were obtained with the second technique, which exhibits a smoothness and more accurate time evolution of the *CDF* of detrended price (Fig. 6).

We extend the analysis of this derivation in the supplementary materials section. The corresponding q, α and D values that fit each of these equations are shown in Figure A.1 (a–c–d). The q and β values were calculated directly after fitting Eq. (A7). The α values are calculated as the power law of the β value, by averaging the power law over a moving time window smaller than the transition zone. The D value is calculated by replacing α in Eq. (A5). The results for q, α and D obtained by applying the least squares method of the CDFs are consistent with the ones calculated focused on the PDFs. The convergence of $q \to 1$ and $\alpha \to 2$ shows that the PDF of x is Gaussian when $t \to \infty$.

Then, we construct a forecast of the S&P500 using the piece-wise function, which components have shown more accurate approximations for the collapse and the expected recovery (Fig. 4a). We note that the stock market trends are approximations based upon the prices for the following trading days affected by breaking news. For that reason, the trend is calculated using $t_0 = 29/02/2020$; the date on which the first death was reported in USA [43]; and $t_f = 23/03/2020$; two months after the first case of contagious was confirmed. Notably, this proposed trend exhibits a better performance than the parabola and the hyperbola.

The uncertainty was modeled with the analytical form of the detrended price by replacing the $\alpha(t)$, $q(t)$, and $D(t)$ calculated in Appendix A into Eq. (4) with their numerical estimates. Once the stochastic term is simulated it is added into the deterministic trend to see the full stochastic path of $I(t)$.

Fig. 7 shows the forecast result identifying the range of possibilities during the market's decline and subsequent recovery. The predicted trend was plotted as a piece-wise function with $t_0 = 29/02/2020$ (Fig. 4). The uncertainty is presented by a shaded contour plot with a scale from 0 to 1. These values represent the probability of a possible variation of $P(X - \overline{X} > |x|)$ which can be visualized as a "cone of uncertainty". This probabilistic path of the S&P500 over the following five months shows the evolution of the PDF of $I(t)$, which diffuses as t increases. Also we see that the real data for index after t_0 (the dashed line) lies within a tiny interval in the vicinity of the predicted trend, mainly in the most probable (dark) regions, verifying that our model is working and is reliable for forecasting with 85% of accuracy.

We construct a forecasting based on essential public information provided by the University of Washington on their web page [29]. A good forecasting is always an iterative process that can be daily updated considering new factors and changes.

6. Conclusions

We have presented a model of the stochastic and systematic risk in the stock market and applied it to forecast the stock market response to COVID-19. We have assumed that the stochastic risk is a q-Gaussian diffusion process. The systemic risk is the deterministic aspect of the market evolution that is often neglected by market analysts but here we have assumed that COVID-19 has a deterministic, exogenous impact on the market. We have dressed the trend with a q-Gaussian diffusion process that we developed in previous work. The response function used for forecasting is a simple behavioral model based on breaking news, which determines the beginning of a decline or a recovery. Our assumption is that markets recover following an exponential function once prices drop to sufficiently low enough levels. This model has been tested on previous financial crises that have occurred in the last 25 years for two indexes; S&P500 and DAX. In future work it can be improved by developing better tools to evaluate systematic risk using behavioral [55] or macroeconomic [56] models. In addition, our stochastic method is still heuristic and it needs to be developed by formulating the governing equations that allows transition from strong- to weak- superdiffusion and converging to a normal diffusion process for large times, as dictated by the classical Central Limit Theorem. However, this method opens up a potential opportunity for risk control in other areas such as climatology, seismology (earthquakes) and communication networks, where large multivariate data sets provide a window on the dynamics of the underlying system.

CRediT authorship contribution statement

Karina Arias-Calluari: Conceptualization, Methodology, Software, Writing - original draft, Validation. **Fernando Alonso-Marroquin:** Writing - review & editing, Investigation, Data curation, Supervision. **Morteza N. Najafi:** Validation, Formal analysis. **Michael Harré:** Investigation, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary analysis

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